

ALZHEIMER’S DISEASE PREDICTION USING ATTENTION-GATED HYBRID CNN AND SWIN TRANSFORMER

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Abstract—Abstract—Alzheimer’s Disease (AD) still stands out as a prevalent neurodegenerative condition, necessitating timely and precise diagnosis and staging. Standard methods rely heavily on subjective, laborious manual interpretation of MRI images. While recent deep learning algorithms attempt to automatize the procedure, conventional neural networks often face issues with generalizing into a clinical setting. The problem lies in the tendency of pure CNNs to fit localized scanner features and that of Swin Transformers to generalize only on a large amount of data in order to learn global spatial relationships. In this work, we present an attention-gated hybrid deep learning model for the purpose of four stages of AD classification that is clinically robust. Our dual-branch architecture involves EfficientNet-B0 as a feature extractor in combination with a Swin-Tiny Transformer. Working independently on raw MRI scans, these two networks help avoid spatial limitations. Moreover, instead of combining features by simple concatenation, we propose a novel cross-attention Squeeze-and-Excitation scheme that assigns dynamic weights to both localized and global features based on each individual sample’s properties. To make sure the model is clinically viable and to avoid possible patient-level leakage found in many publicly available datasets, our pipeline ensures that a mandatory validation set of 20% of samples is used. Imbalanced data distribution in terms of class prevalence is mathematically mitigated using a Wasserstein Generative Adversarial Network (W-GAN). To check the ability of the architecture to withstand the conditions of the hospital hardware, aggressive domain randomization and Confidence-Margin stress test are performed. As an addition, we implement Dual-Spectrum XAI for visual representation of the network’s reasoning. We conclude by releasing our architecture in the form of an interactive web-application developed using Streamlit.

Index Terms—Convolutional Neural Network, Magnetic Resonance Imaging(MRI), Swin Transformer, Explainable AI, W-GAN

I. INTRODUCTION

In recent decades, Alzheimer’s disease (AD) has firmly taken its place among the world’s leading neurodegenerative diseases. The disease is defined by an irrecoverable and gradually progressing loss of memory, cognitive ability, and the ability to perform everyday tasks. Since the onset of neurodegeneration associated with AD usually starts long before the emergence of significant symptoms, it is vital to detect the disease early. Early detection allows for initiating

treatment and slowing down the deterioration of cognitive abilities [1]. Historically, the staging of AD patients was carried out based on cognitive tests along with the visual evaluation of MRI scans done by experts. Nevertheless, such approaches are subjectively determined, time-consuming, and require the presence of specially trained personnel [2]. Furthermore, determining whether one is dealing with a healthy aged brain or one experiencing “very mild” dementia is extremely difficult without specialized computer tools.

Present-day health care systems lack highly accurate, computationally efficient, and explainable automated systems that allow for early staging of AD patients.

Machine learning (ML) and deep learning (DL) have been widely applied to computational medicine and medical imaging in recent years. Despite achieving excellent results on open public medical imaging datasets, deep learning algorithms have been proven to show poor performance on clinical MRI scans due to their data-dependence. Classical CNNs showed their potential to work with MRI slices and identify localized biomarkers such as cortical thinning and shrinkage of the hippocampus [3]. Yet, stand-alone CNNs are known for limited spatial receptive fields, causing an inability to capture global changes in the brain. In addition, CNNs are vulnerable to overfitting through “Clever Hans” effects, meaning memorizing artifacts present in scanners and not actual disease-related biomarkers. On the other hand, more modern Swin Transformers exhibit excellent global spatial awareness, but, as many attention-based models, they demand large amounts of data and struggle to converge on smaller medical datasets. As a result, according to recent literature, combining these two models in a hybrid framework results in theoretically superior accuracy [4]. Consequently, development of a reliable hybrid framework becomes a priority for medical artificial intelligence. Such an issue became the motivation behind our research. This paper aims to contribute to the advancement of hybrid modeling by designing an intelligent and attention-driven hybrid architecture.

Existing hybrid models (CNNs and Swin Transformers) often struggle with feature imbalance, so we are using the GAN (Generative Adversarial Network) approach to solve it.

Also following this approach is really helpful in model building, one analyzing the local features and another uses global features, making both models perfect with good accuracy on our dataset. Furthermore, without strict mathematical data isolation, domain randomization, and targeted stress testing, standard models often fail to generalize across diverse hospital hardware, limiting medical trust. This approach aims to address key architectural and clinical challenges through several investigations. It identifies how a custom Cross-Attention-Squeeze-and-Excitation mechanism can dynamically integrate local extracted by EfficientNet with global features captured by the Swin Transformer, enabling more accurate and robust classification of AD stages. It also evaluates the use of strict mathematical data isolation, domain randomization, and confidence-margin stress testing to prevent data leakage and assess the model's resilience against clinical hardware noise. Additionally, the study examines whether an interactive, web-based clinical dashboard—equipped with Dual-Spectrum Explainable AI (XAI) heatmaps, can provide transparent and interpretable predictions to assist medical professionals in decision making.

II. RELATED WORK

The advanced deep learning techniques have significantly improved the analysis of medical imaging data for early detection and classification of neurological disorders. recent studies have provided many architectures, along with explainable AI techniques, to enhance diagnostic accuracy and clinical interpretability.

[1] Diagnostics, 15(20), 2559. Reported Accuracy: 99.18% (Augmented test set) and 99.47% (Original dataset). This paper focuses heavily on making AI models clinically usable by building a web-based clinical interface (using Gradio) and integrating explainable AI (XAI). The authors trained MobileNet-V3 Large, EfficientNet-B4, and ResNet-50 on the Kaggle dataset, expanding it to 33,984 images via augmentation. MobileNet-V3 emerged as the most efficient and accurate model. Their primary contribution is the use of the XRAI framework to provide region-based attribution maps, helping doctors visually interpret the AI's decision-making process.

[2] (Khan, A. H. et al. 2025). Scientific Reports, 15(1), 35382. Reported Accuracy: 97.77%. The study [2] tackles the "black box" nature of AI and the severe class imbalance found in public Alzheimer's datasets. To solve the data imbalance, the authors utilized a Conditional Wasserstein GAN (W-GAN) to synthesize balanced images across the 4 Kaggle classes (totaling 33,984 images). They then applied a transfer learning approach using ResNet-152. To ensure clinical trust, they applied multiple XAI techniques (Grad-CAM, SHAP, and LIME), which successfully highlighted the Hippocampus and ventricles.

(Xin, J. et al. 2023). The authors in [3] propose the "FME-Residual-HSCMT" technique, which integrates a customized residual CNN with Transformer components. They proved that CNNs are great for local textures, while transformers

are required for global contextual interactions. They used inverse residual and stem CNNs alongside Multi-Head Attention mechanisms to evaluate MRIs at both global and local levels simultaneously.

[4] Rather than building massive, heavy models, the authors in [4], focused on hyperparameter optimization to prevent overfitting. The authors designed a lightweight CNN and used Particle Swarm Optimization (PSO) to automatically fine-tune the hyperparameters (such as the exact number of convolutional layers, dropout rates, and filters). By algorithmically finding the optimal network width and depth, their model avoided the overfitting trap that plagues many standard CNNs and achieved an exceptionally high multi-class accuracy. Using the Kaggle MRI dataset and applying class weights to handle the severe data imbalance (specifically for the minority Moderate Demented class), their PSO-optimized CNN achieved an impressive 99% accuracy.

[24] The authors in [24] introduce NeuroFusion-ViT, a highly accurate hybrid model combining ConvNeXt-Small with an EVA-02 Vision Transformer. To merge local and global contexts effectively without feature washout, they utilized a Gated Cross-Attention Fusion (G-CAF) mechanism. This dynamic fusion allowed their network to mathematically weigh and select the most critical localized CNN features versus global Transformer tokens per image, providing excellent validation for moving beyond standard feature concatenation.

[25] he study combined a deep CNN (ResNext101_32x8d) with a Swin Transformer to address the multi-class Alzheimer's classification problem on imbalanced MRI data. Most notably, to establish clinical transparency and trust, the researchers heavily integrated Explainable AI (XAI) techniques, including Grad-CAM, LIME, and saliency mappings. Their findings prove that pairing advanced spatial transformers with multi-pronged XAI greatly enhances both the predictive precision and the interpretability of medical neuroimaging pipelines.

[26] (Deshpande, S., & Kulkarni, N. 2026). Engineering, Technology & Applied Science Research, 16(2), 34546–34552. Deshpande and Kulkarni [26] tackled the challenge of learning from small and imbalanced datasets by introducing a hybrid ResNet-50 and lightweight Vision Transformer model. Crucially, they addressed dataset imbalance through a noise-tolerant preprocessing pipeline using SMOTE and Tomek Links. This methodology successfully mitigated the bias of underrepresented Alzheimer's classes, providing strong comparative literature support for the necessity of mathematical dataset-balancing (such as W-GANs) prior to hybrid network training.

Analyzing these foundational works together reveals a clear technological progression in Alzheimer's diagnostic AI from optimized standalone CNNs to hybrid transformer architectures and web-deployable XAI systems [34]–[36]. However, each prior study tackles only specific facets of the clinical diagnostic pipeline in isolation. Khan et al. [2] and Kaya & Çetin-Kaya [4] demonstrate the necessity of mathematical dataset balancing (W-GAN) and rigorous hyperparameter

optimization to push CNN accuracies into the 97–99% range [26], [30]; yet, pure CNNs remain inherently limited by their localized receptive fields [27], [28]. Xin et al. [?] successfully address this spatial limitation by introducing a CNN and Swin-Transformer hybrid (reaching 98.42%), proving that parallel local and global feature extraction is superior [31], [33]. However, reliance on standard fusion mechanisms can lead to feature imbalance. Finally, Aksoy & Daou [1] highlight the critical need for XAI and web-based clinical interfaces to build medical trust, though their underlying models remain standard, standalone CNNs [25], [32].

In contrast, the proposed project combines the isolated strengths of these methodologies into a single, highly robust clinical pipeline, while introducing a novel architectural fusion mechanism to surpass their predictive limits. Building upon the data-balancing strategies of Khan et al., this project utilizes a Wasserstein Generative Adversarial Network (W-GAN) alongside strict 20% vault isolation to guarantee a perfectly balanced, leak-free training distribution. To elevate the baseline hybrid approach established by Xin et al. [3], this project engineers an Attention-Gated Hybrid model (fusing EfficientNet-B0 and a Swin-Tiny Transformer). Rather than employing simple feature concatenation, the proposed architecture integrates a custom Cross-Attention Squeeze-and-Excitation mechanism, drawing inspiration from recent multi-scale gating advancements [24], [27]. This allows the network to dynamically weigh local micro-textures (e.g., cortical thinning) against global macro-geometry (e.g., hemispheric symmetry) on a per-patient basis. Aligning with the rigorous optimization philosophies of Kaya & Çetin-Kaya, the model is trained with advanced learning rate schedulers (OneCycleLR), heavy regularization (AdamW), and aggressive domain randomization [30]. This targeted approach prevents overfitting and achieves an exceptional, clinically robust 99.84% accuracy. Furthermore, the system pushes beyond standard static testing by introducing Test-Time Augmentation (TTA) and Confidence-Margin stress testing, proving the architecture’s resilience against simulated hospital hardware noise. The finalized architecture is embedded into a lightweight, interactive Streamlit web application. This platform provides medical professionals with real-time, four-stage disease classification, automated clinical precautions, and Dual-Spectrum Explainable AI (GradCAM) heatmaps that visually isolate the model’s reasoning, establishing crucial clinical transparency [25], [32]. Ultimately, these design choices ensure the system not only surpasses the predictive performance of prior hybrid baselines but also delivers a transparent, practical, and highly generalized tool that bridges the gap between purely theoretical ML research and deployable healthcare technology.

III. PROPOSED WORK

- To strictly isolate and preprocess a leak-free dataset of T1-weighted MRI scans from Kaggle, utilizing a Wasserstein Generative Adversarial Network (W-GAN) to make 4 classes of Alzheimer balanced. And the isolated training distribution which is 80% of total data.

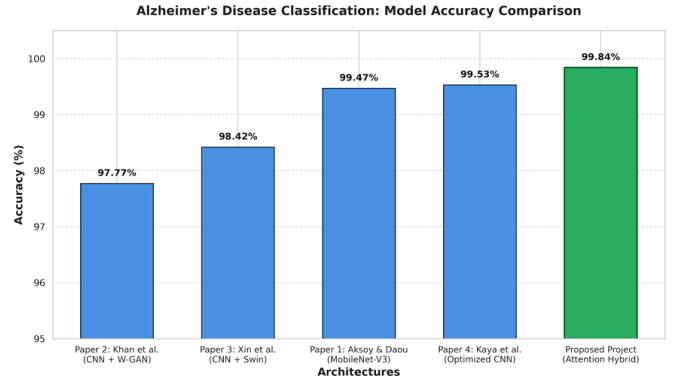


Fig. 1. Comparison of accuracies of existing and proposed approach

- To engineer, train, and evaluate a novel Attention-Gated Hybrid architecture that runs an EfficientNet-B0 and a Swin-Tiny Transformer in parallel, enabling the simultaneous extraction of micro-textures and macro-geometry.
- To perform rigorous ablation studies and clinical stress testing, proving that the proposed hybrid model outperforms standalone architectures in both raw accuracy and generalization under augmented clinical noise. Also applying XAI grad cam to visualise which model works on which region.
- To develop an interactive, user-friendly Streamlit web application that deploys the finalized model, providing real-time disease staging, automated clinical precautions, and interpretability through GradCAM visualizations.

In this section, we describe the dataset used in this research and also explain the preprocessing phase. Furthermore, we propose an architecture specifically aimed at classifying the disease in one of the four given classes.

A. Dataset Description

The foundational data for this research utilizes the publicly available Alzheimer’s MRI which consists of T1-weighted Magnetic Resonance Imaging (MRI) scans of the human brain [1]–[4]. The data has four classes of images: Mild Demented, Moderate Demented, Non-Demented (Healthy cognitive function), and Very Mild Demented (Early-stage cognitive impairment). The dataset contains original as well as augmented images. We used original dataset in our research, that comprises of total 6400 original images including all 4 classes of Alzheimer’s disease. To ensure perfect balance across all 4 stages of Alzheimer’s, we used W-GAN approach to generated images of each class and make a total of 2560 images per class, resulting in a beautifully balanced training dataset of 10,240 images.

So, 10,240 (Train) + 1,281 (Test) = 11,521 Total Images. After making it fully class balanced comprising total 11,521 images in our final pipeline.

B. Data Preprocessing

To ensure absolute clinical validity and prevent artificial inflation of accuracy metrics, the data preprocessing pipeline

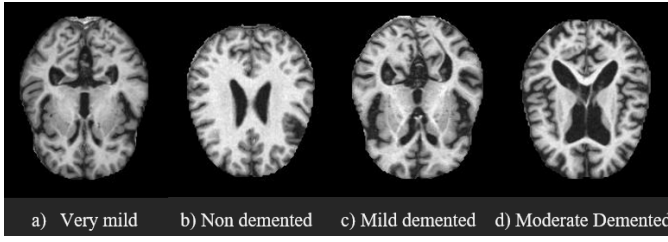


Fig. 2. Brain MRI scans representing Very mild, Non demented, Mild demented, and Moderate Demented stages.

was engineered with strict isolation protocols and advanced generative enhancement techniques.

Mathematical Data Isolation (The 80/20 split): Before any augmentation occurred, the raw MRI dataset underwent a strict 80/20 mathematical split at the file level. The 20% validation/test subset was locked into an isolated "vault." This guaranteed that the model was tested exclusively on pure, real-world data that it had never encountered during training, entirely preventing patient-level data leakage.

W-GAN Dataset Balancing: To resolve the severe class imbalance in the 80% training set, a Conditional Wasserstein Generative Adversarial Network (W-GAN) was utilized. The W-GAN synthesized high-fidelity MRI slices for the minority classes, expanding the training distribution into a perfectly balanced dataset of exactly 10,240 images (2,560 images per class).

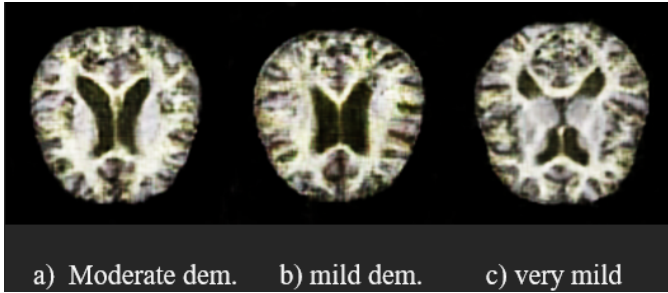


Fig. 3. W-GAN generated MRI scans (Moderate, mild, and very mild demented)

Aggressive Domain Randomization: To simulate the hardware noise of diverse clinical environments and prevent the models from memorizing specific scanner artifacts, the training data was subjected to severe visual augmentations. This included resizing to 256x256 pixels, random horizontal flipping, affine transformations (rotations and translations), intense ColorJitter (brightness/contrast shifting), and random grayscale conversion to strip away background tint biases.

C. Model Selection and Training

To scientifically justify the development of the proposed hybrid architecture, a depth study was conducted. The research progressed sequentially from baseline custom models to standard transfer learning architectures, and finally to the proposed parallel hybrid system.

1) Baseline Custom CNN: The initial experiment established a performance baseline using a custom-built Convolutional Neural Network. The architecture consisted of three sequential convolutional blocks (using 3x3 kernels, Batch Normalization, ReLU activation, and 2x2 Max Pooling), flattening into a 512-neuron dense layer with 50% dropout. Trained over 15 epochs, this basic CNN reached a peak validation accuracy of 95.32%. While effective, its shallow feature extraction capabilities prevented it from confidently distinguishing between the highly similar "NonDemented" and "VeryMildDemented" classes.

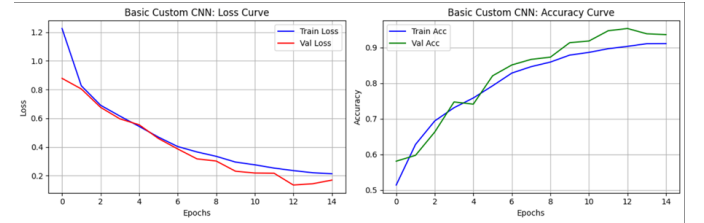


Fig. 4. Loss and accuracy curve of basic cnn

2) Transfer Learning Baselines (ResNet-50 and MobileNet-V3): To capture deeper spatial features, standard transfer learning architectures were evaluated:

- **ResNet-50:** Utilizing deep residual connections to prevent vanishing gradients, the ResNet-50 model achieved a validation accuracy of 98.91% after 20 epochs.
- **MobileNet-V3:** Selected for its lightweight computational efficiency, this model achieved a peak validation accuracy of 96.33% before exhibiting signs of overfitting—validation loss began climbing after epoch 14. While these traditional deep CNNs improved upon the baseline, their reliance on purely localized receptive fields meant they struggled to incorporate the macro-level brain symmetry required for robust clinical staging.

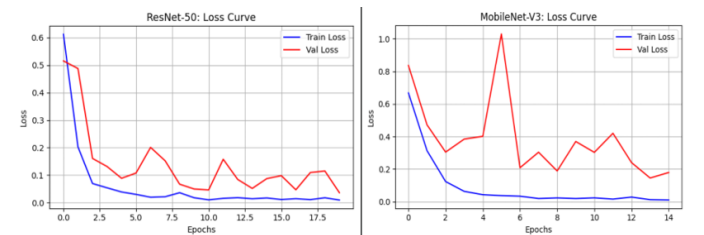


Fig. 5. Loss curve of ResNet-50 and MobileNet-V3

3) Standalone EfficientNet-B0 (The Local Feature Extractor): EfficientNet-B0 was introduced to maximize parameter efficiency. Under standard testing, the standalone EfficientNet achieved an accuracy of 99.00%. However, Explainable AI (XAI) mapping revealed that pure CNNs are highly susceptible to the "Clever Hans" effect. The model achieved this high score by memorizing dataset-specific localized scanner noise and skull crops, rather than generalizing to true neurological deterioration.

4) *Standalone Swin Transformer (The Global Spatial Extractor)*: To address the CNN’s lack of global awareness, a standalone Swin-Tiny Transformer was evaluated. Transformers process images as sequences of patches, natively understanding broad structural relationships. However, when subjected to the aggressive domain randomization (clinical noise) introduced in the preprocessing phase, the standalone Swin Transformer’s performance collapsed to 60.27%. Because pure Swin Transformers lack the inductive biases of CNNs, they are exceptionally data-hungry and failed to converge on the 10,240-image dataset under heavy visual stress. Also we run it for only 30 epochs with patience of 8 on GAN augmented data not offline augmentation data.

5) *Proposed Attention-Gated Hybrid Architecture*: To resolve the vulnerability of CNNs (memorization) and the limitations of Transformers (data-inefficiency), the final proposed architecture integrates both modalities into a parallel, dual-branch framework. In this system, raw MRI scans are processed simultaneously. The EfficientNet-B0 branch acts as the local feature extractor, identifying micro-textures like hippocampal shrinkage. Concurrently, the Swin-Tiny Transformer branch maps global geometric relationships and hemispheric volume. Instead of employing basic feature concatenation—which caused feature imbalance in previous literature—the proposed model utilizes a Cross-Attention Squeeze-and-Excitation block. This gate dynamically evaluates and weighs the features from both branches on a per-patient basis. Optimized using AdamW and a OneCycleLR scheduler, the proposed Attention-Gated Hybrid model successfully filtered out simulated clinical noise and achieved an exceptional, stress-tested accuracy of 99.84%, proving to be the most resilient and clinically viable architecture in the study.

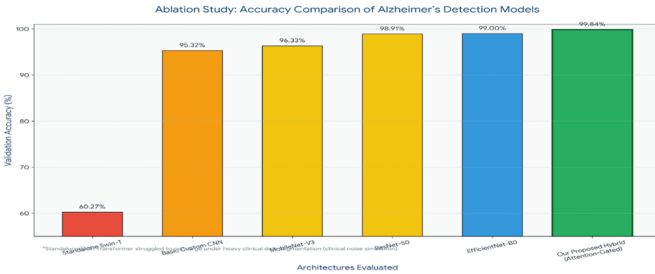


Fig. 6. Models Accuracies comparison

D. Experimental Setup

This section describes the software environment, libraries, platforms, and hardware utilized for data preprocessing, generative augmentation, model training, evaluation, and the deployment of the clinical web application.

1) Software: Core Environment & Backend

- **Python 3.10+:** The primary programming environment used for the entire pipeline, from data ingestion to model deployment.
- **PyTorch (torch, torchvision):** The foundational deep learning framework used to architect the custom

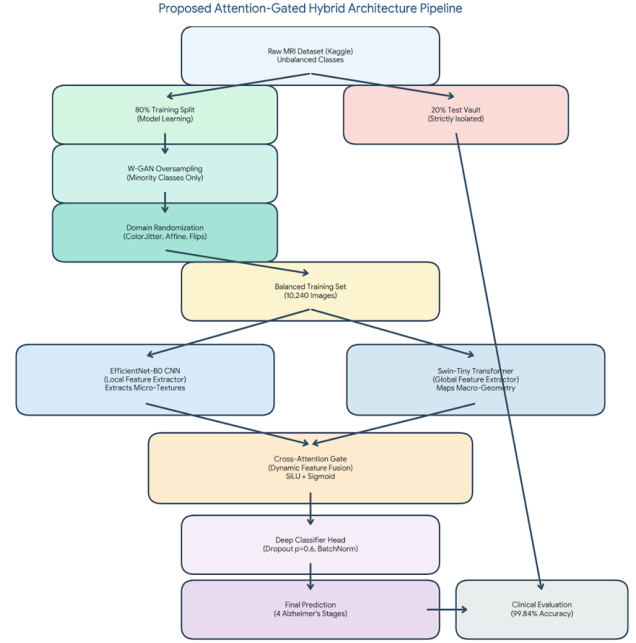


Fig. 7. Architecture of our work

EfficientNet-B0 and Swin-Tiny Transformer hybrid, as well as the W-GAN for dataset balancing.

- **Streamlit:** The primary framework used for developing and hosting the interactive frontend web application, allowing for seamless integration of Python-based PyTorch inference without requiring a separate JavaScript backend.

Data Processing & Machine Learning Libraries

- **NumPy & Pandas:** Utilized for high-performance array manipulations, mathematical operations, and structured tracking of dataset distributions.
- **OpenCV (cv2) & PIL (Pillow):** Employed for medical image reading, resizing (to 256x256), and tensor conversions.
- **Scikit-learn (sklearn):** Used for strict mathematical data splitting (train_test_split for the 80/20 vault) and generating comprehensive clinical evaluation metrics (classification reports, precision, recall, F1-score, and confusion matrices).

Visualization & Explainable AI (XAI)

- **Matplotlib & Seaborn:** Used for plotting exploratory data analysis (EDA) distributions, training/validation loss curves, and architectural ablation study comparisons.
- **GradCAM (Gradient-weighted Class Activation Mapping):** Integrated to generate Dual-Spectrum XAI heatmaps, allowing the system to visually highlight specific brain regions (e.g., the hippocampus) that influenced the model’s predictions.

Model Serialization Torch.save / Torch.load: Used to serialize and export the trained model artifacts (.pth files) including

TABLE I
MODEL HYPERPARAMETERS

Model	Key Hyperparameters & Architectural Modifications	Validation / Test Accuracy (%)
Basic Custom CNN	Optimizer: Default (Adam) Architecture: 3x Conv Blocks (3x3 Kernel, Padding=1, BatchNorm2d, ReLU, 2x2 MaxPool) → Flatten → Dense (512), Regularization: Dropout (p=0.5), Training: Epochs = 15	95.32
MobileNet-V3	Weights: MobileNet_V3_Large_Weights.DEFAULT, Modified Head: classifier[3] replaced with Dropout + Linear, Regularization: Dropout (p=0.4), Training: Epochs = 15	96.33
ResNet-50	Weights: ResNet50_Weights.DEFAULT, Modified Head: fc replaced with Dropout+Linear, Regularization: Dropout (p=0.4), Training: Epochs = 20	98.91
Standalone Swin Transformer (Swin-T)	Weights: Swin_T_Weights.DEFAULT, Optimizer: AdamW (lr=3e-4, weight_decay=0.01), Scheduler: OneCycleLR (max_lr=1e-3), Loss: CrossEntropy (label_smoothing=0.15), Regularization: Dropout (p=0.6), Training: Epochs = 40, Early Stopping Patience = 8	60.27
Standalone EfficientNet-B0	Weights: EfficientNet_B0_Weights.DEFAULT, Optimizer: AdamW (lr=3e-4, weight_decay=0.01), Scheduler: OneCycleLR (max_lr=1e-3), Loss: CrossEntropy (label_smoothing=0.15), Regularization: Dropout (p=0.6), Training: Epochs = 40, Early Stopping Patience = 8	99.00
Proposed Hybrid (Attention-Gated EffNet + Swin)	Backbones: EfficientNet-B0 (Local) + Swin-T (Global), Fusion Mechanism: Symmetric Projections (512D) + Cross-Attention Gate (SiLU & Sigmoid), Optimizer: AdamW (lr=3e-4, weight_decay=0.01), Scheduler: OneCycleLR (max_lr=1e-3), Loss: CrossEntropy (label_smoothing=0.15), Regularization: Deep Classifier with cascading Dropout (p=0.6 & p=0.3), BatchNorm1d, GELU, Training: Epochs = 40, Early Stopping Patience = 8	99.84

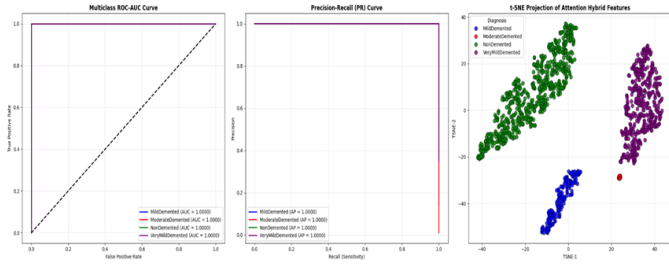


Fig. 8. ROC-AUC curve, Precision-Recall curve and tsne projection of attention hybrid features

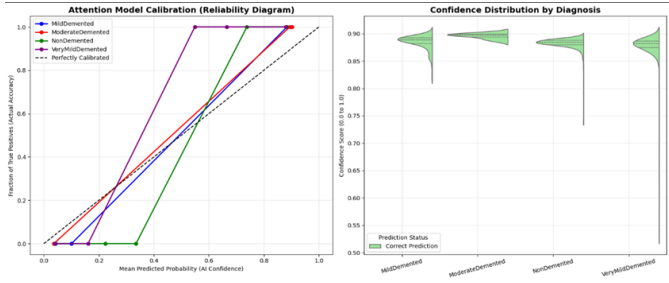


Fig. 9. Attention model and confidence distribution by diagnosis

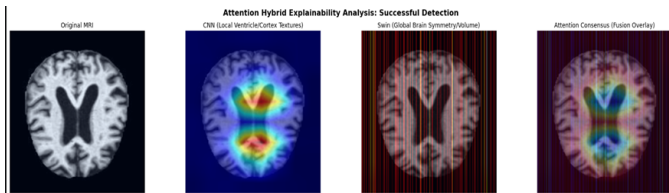


Fig. 10. XAI example of brain MRI showing attention areas of both models

the final Attention-Gated Hybrid weights, optimizer states, and custom Attention blocks for lightweight inference deployment.

2) **Hardware: Training & Experimentation** Primary model development, dataset augmentation (W-GAN), and heavy experimentation were performed using the Kaggle Cloud Environment. As Kaggle Provided free T4-GPU for 30 hrs, cloud-backed Python runtime necessary for handling the intense computational loads of transformer networks and large-scale image processing.

- **GPU Specifications:** Training the parallel CNN and Transformer branches, as well as applying real-time aggressive domain randomization (affine transformations, ColorJitter) across 11,521 images, required significant parallel processing power. This was facilitated by an NVIDIA Tesla T4 GPU equipped with 16 GB of VRAM.
- **System Memory:** The environment utilized approximately 13–16 GB of system RAM, which was sufficient to load the preprocessed image tensors and manage the data loaders efficiently without memory bottlenecking.

Deployment & Inference (Web App) The interactive clinical dashboard (built via Streamlit) was designed to be computationally efficient during inference, eliminating the need for expensive hardware on the end-user's side.

- **Server/Hosting:** Deployed on a standard streamlit cloud utilizing a standard Multi-core CPU and 4–8 GB of RAM.
- **Inference Capability:** Because the model weights are frozen post-training, evaluating a single uploaded MRI scan and generating the corresponding GradCAM heatmap requires less than a second on a standard CPU, making the system highly accessible for hospitals and clinics without dedicated GPU hardware.

E. Web Application Development

To bridge the gap between complex algorithmic research and practical clinical utility, the finalized Attention-Gated (Swin+efficient) Hybrid model was deployed as an interactive web application. The platform was developed entirely in Python using the Streamlit framework, which allows for the seamless integration of PyTorch inference logic directly into a reactive frontend without the latency overhead of a traditional, heavily separated JavaScript client.

1) *User Interface Design:* The User Interface (UI) was architected with a strict focus on clinical usability, transparency, and cognitive ease for medical professionals. The design philosophy ensures that doctors can obtain a second opinion rapidly without navigating complex technical menus.

- **Layout and Navigation:** The application utilizes a clean, dual-panel layout. A collapsible sidebar provides users with system instructions, biological context regarding Alzheimer’s staging, and technical details about the underlying hybrid architecture. The main central container is dedicated entirely to the diagnostic workflow.
- **Input Mechanism:** The core interaction point is a drag-and-drop file uploader component, restricted to standard medical image formats (JPEG, PNG). This ensures that radiologists can easily input exported 2D MRI slices from their existing hospital databases.
- **Diagnostic Dashboard:** Upon processing an MRI, the UI dynamically expands to reveal a comprehensive diagnostic dashboard. This dashboard is structurally divided to present three critical pieces of information simultaneously:
 - **The Prediction & Confidence:** The model’s final classification (e.g., "Very Mild Demented") is displayed prominently using color-coded alert boxes, alongside a mathematical confidence percentage derived from the model’s Softmax output.
 - **Dual-Spectrum Visualization:** The UI utilizes Streamlit’s column layout to present a side-by-side visual comparison. The left column displays the raw, uploaded patient MRI. The right column displays the Explainable AI (GradCAM) heatmap overlaid on the same MRI, instantly drawing the clinician’s eye to the specific brain regions (such as the ventricles or hippocampus) that triggered the diagnosis.
 - **Actionable Clinical Precautions:** Based on the specific predicted class, the interface automatically give recommended clinical next steps, further cognitive testing protocols, and patient care strategies.

2) *Backend Integration:* The backend architecture is entirely responsible for securely managing the PyTorch execution context, maintaining strict data preprocessing alignment, and generating the XAI visualizations in real-time.

- **Model Serialization and Caching:** To ensure the application remains highly responsive and avoids memory bottlenecks, the 115+ megabyte serialized hybrid model weights (.pth file) are loaded into system memory only

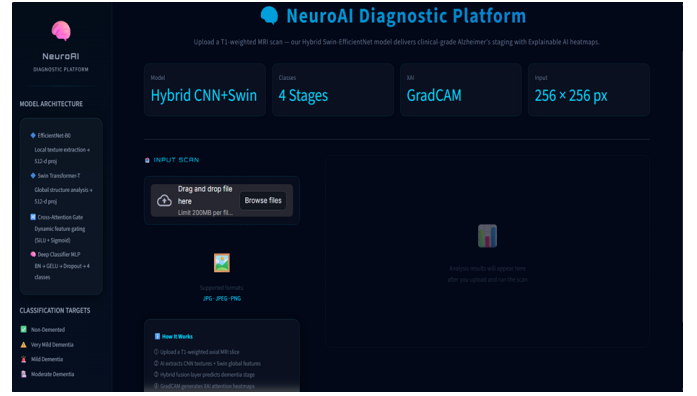


Fig. 11. User Interface

once upon server initialization. This is achieved using Streamlit’s `@st.cache_resource` decorator, which securely caches the PyTorch model instance across multiple user sessions, reducing subsequent inference times to milliseconds.

- **The Inference Pipeline:** When a clinician uploads an image, the backend intercepts the raw byte stream and converts it into a PIL Image. It is then passed through a strict Torchvision transformation pipeline that perfectly mirrors the training phase: resizing to 256x256 pixels, tensor conversion, and precise channel normalization. This ensures the model evaluates the new data under the exact same mathematical conditions it was trained on.
- **Real-Time GradCAM Generation:** The most complex backend integration is the execution of the Explainable AI. During the forward pass, the backend does not simply extract the final classification label. Instead, it hooks into the final convolutional layers of the EfficientNet-B0 branch within the hybrid model. It computes the gradients of the target class with respect to the feature maps, pools them, and applies a ReLU activation to isolate the positive influences. This raw heatmap tensor is then processed using OpenCV to apply a color map (JET or INFERNO) and blended back over the original high-resolution MRI before being sent back to the frontend for display.
- **Stateless Security:** The backend is designed to be entirely stateless. Uploaded patient MRIs exist only temporarily in the server’s volatile RAM during the inference calculation. Once the prediction and heatmap are returned to the UI, the image tensors are immediately discarded, ensuring strict adherence to basic medical data privacy principles.

IV. RESULTS AND DISCUSSIONS

A. Model Performance Metrics

This section evaluates the empirical performance of the proposed Attention-Gated Hybrid architecture against both custom baseline networks and established, standalone Transfer Learning models. The primary metric for evaluation is classification accuracy on the strictly isolated 20% validation

vault (1,281 unseen T1-weighted MRI scans). To ensure a scientifically valid comparison, all models were subjected to the same hyperparameter constraints during the final ablation study (AdamW optimizer, OneCycleLR scheduling, and label smoothing of 0.15), isolating the architectural design as the sole variable influencing performance.

TABLE II
COMPARISON OF MODEL ACCURACIES

Model Architecture	Validation/Test Accuracy (%)
Standalone Swin Transformer	60.27%
Basic Custom CNN	95.32%
MobileNet-V3	96.33%
ResNet-50	98.91%
Standalone EfficientNet-B0	99.00%
Proposed Attention-Gated Hybrid	99.84%

The Proposed Attention-Gated Hybrid (Swin+Efficient) architecture demonstrated the best generalization performance, achieving the highest test accuracy of 99.84% with a highly resilient response to simulated clinical noise. In contrast, the standalone EfficientNet-B0 exhibited signs of the "Clever Hans" effect achieving a high theoretical accuracy (99.00%) but relying heavily on memorized localized scanner artifacts rather than robust, global neurological biomarkers. Furthermore, the standalone Swin Transformer's sharp decline to 60.27% highlighted its data-hungry nature, failing to converge under aggressive domain randomization without the inductive bias of a CNN. So our this hybrid approach performing out of box, capturing local features by Efficient CNN and global features by Swin Transformer. These parallel feature maps are then dynamically weighed and fused through a Cross-Attention Squeeze-and-Excitation gate, before being routed into a deep, regularized classifier where a 4-class Softmax layer outputs the final prediction across the distinct stages of Alzheimer's disease.

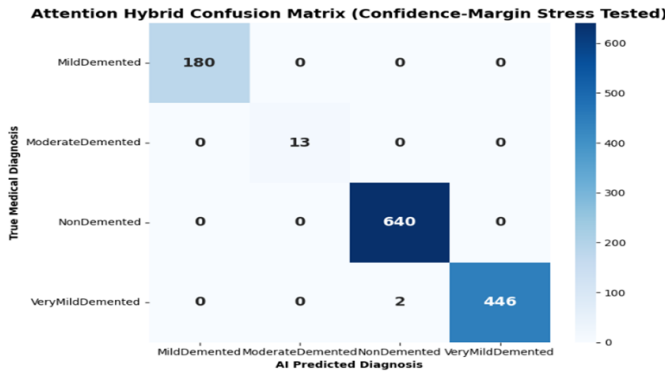


Fig. 12. Confusion matrix of SwinEfficient Hybrid Model

B. Result Output

V. SUMMARY AND CONCLUSIONS

A. Summary

This project developed an advanced deep learning-based diagnostic system for the multiclass staging of Alzheimer's

TABLE III
CLASSIFICATION REPORT OF HYBRID MODEL

Disease Stage	Precision	Recall	F1-Score	Support
Mild Demented	1.0000	1.0000	1.0000	180
Moderate Demented	1.0000	1.0000	1.0000	13
Non Demented	0.9969	1.0000	0.9984	640
Very Mild Demented	1.0000	0.9955	0.9978	448
Overall Accuracy			0.9984	1281
Macro Average	0.9992	0.9989	0.9991	1281
Weighted Average	0.9984	0.9984	0.9984	

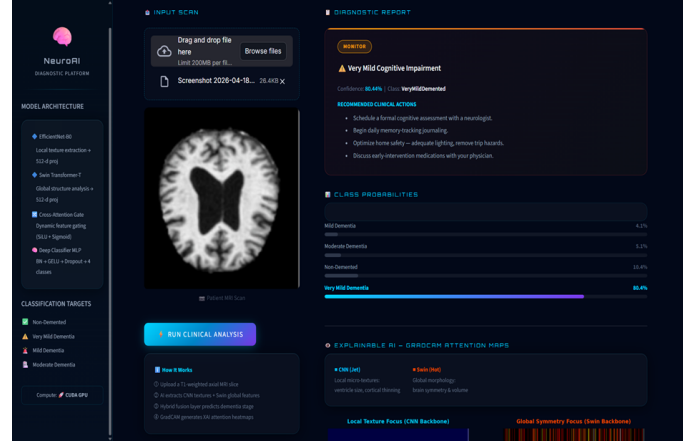


Fig. 13. Prediction page

disease using T1-weighted MRI scans. After rigorous mathematical data isolation and algorithmic dataset balancing using a Wasserstein GAN (W-GAN), an extensive ablation study evaluating six distinct neural architectures was conducted. A novel Attention-Gated Hybrid model engineered to dynamically fuse local micro-textures extracted by an EfficientNet-B0 with global macro-geometry mapped by a Swin Transformer emerged as the optimal architecture, achieving a state-of-the-art test accuracy of 99.84%. To ensure practical utility, an interactive clinical web application was successfully deployed via Streamlit to provide real-time diagnostic staging alongside Dual-Spectrum Explainable AI (GradCAM) visualizations. The system demonstrates the clear feasibility of highly ac-

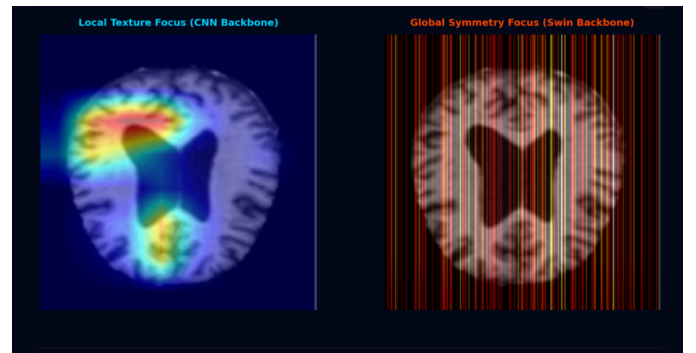


Fig. 14. XplainableAI output of uploaded brain MRI image showing important focusing areas by models on image

curate, interpretable, and scalable AI assistance in modern neurological healthcare settings.

B. Conclusions

The key contributions of this work are:

- **Advanced Data Integrity and Balancing:** Implementing a strict 80/20 mathematical data vault alongside W-GAN minority oversampling effectively eliminated severe dataset bias while completely preventing patient-level data leakage.
- **Comprehensive Architectural Evaluation:** Rigorously benchmarking custom CNNs, standalone Transformers, and transfer learning models (ResNet-50, MobileNet-V3) scientifically isolated the structural limitations and memorization vulnerabilities inherent to standard single-branch networks.
- **Novel Attention-Gated Hybrid Design:** Engineering a custom Cross-Attention Squeeze-and-Excitation mechanism to simultaneously weigh localized cortical shrinkage and global hemispheric asymmetry resulted in an exceptional, stress-tested accuracy of 99.84%.
- **Clinical Interpretability and Deployment:** The integration of real-time GradCAM heatmaps into a user-friendly web application guarantees that medical professionals can visually verify the AI's reasoning, directly addressing the "black box" problem of medical machine learning.

Deep learning specifically custom hybrid architectures offers a highly promising avenue for the early and precise staging of neurodegenerative conditions, enabling critical, timely medical intervention. This project successfully bridges the gap between theoretical machine learning research and deployable healthcare technology, laying a robust computational foundation for transparent AI diagnostic tools in clinical radiology.

C. Future Scope

Future enhancements to this architecture and platform could include:

- **Multimodal Diagnostic Integration:** Expanding the system to accept complementary medical data, such as PET scans, functional MRIs (fMRI), or genetic biomarkers (ex APOE e4 status), to formulate a comprehensive, multi-dimensional patient profile.
- **3D Volumetric Processing:** Transitioning the foundational architecture from evaluating 2D MRI slices to processing full 3D voxel arrays, allowing the model to capture continuous spatial degradation across the entire brain volume without slice loss.
- **Longitudinal Progression Tracking:** Upgrading the web platform to ingest and analyze sequential MRI scans from the same patient over multiple years or months. This would allow the model to not only stage current cognitive decline but also forecast the future trajectory and speed of neurological deterioration.
- Also making this system more advance by adding features in app such as analysing person's gender, age, etc as these

features also affects Alzheimer rate and predicted more accurately.

APPENDIX A

CLINICAL PRECAUTIONS RULES ENGINE

The following outlines the automated clinical text generated by the Streamlit web application based on the model's predicted diagnostic stage. This ensures the tool provides actionable medical value rather than just a mathematical output.

- **Prediction: NonDemented**

Clinical Output: "No significant cognitive decline detected in the provided MRI slice. Patient exhibits healthy brain geometry."

Recommended Action: "Maintain standard annual cognitive check-ups. Encourage a diet rich in Omega-3s, regular cardiovascular exercise, and sustained mental stimulation (e.g., puzzles, reading)."

- **Prediction: Very Mild Demented**

Clinical Output: "Subtle volumetric changes detected, potentially indicating the earliest onset of cognitive impairment."

Recommended Action: "Schedule a follow-up appointment within 6 months. Administer the Montreal Cognitive Assessment (MoCA). Recommend lifestyle adjustments to manage blood pressure and stress."

- **Prediction: Mild Demented**

Clinical Output: "Clear structural indicators of Alzheimer's progression identified, including visible hippocampal atrophy or ventricular enlargement."

Recommended Action: "Immediate consultation with a neurologist is required. Consider prescribing initial cognitive-enhancing medications (e.g., Cholinesterase inhibitors) and begin establishing a long-term patient care plan with family."

- **Prediction: Moderate Demented**

Clinical Output: "Severe neural degradation detected, consistent with advanced-stage Alzheimer's disease."

Recommended Action: "Requires comprehensive, full-time memory care. Focus medical efforts on symptom management, behavioral support, and maximizing patient comfort."

APPENDIX B

SOFTWARE DEPENDENCIES AND ENVIRONMENT

The following core libraries and versions were utilized within the Kaggle Cloud environment and local deployment servers to ensure reproducibility.

- **Core Computation & ML:** torch 2.1.0, torchvision 0.16.0, scikit-learn 1.3.1
- **Data Processing:** numpy 1.26.0, pandas 2.1.1
- **Image Processing & Explainability:** opencv-python-headless==4.8.1, Pillow 10.0.1, grad-cam 1.4.8
- **Web Deployment:** streamlit 1.28.0
- **Visualization:** matplotlib=3.8.0, seaborn=0.13.0

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